Optical Spectroscopy — Session 01 (Master Protocol)

Document 1 of 3 — Shimadzu UV-2550 UV/Vis Spectrophotometer

UNR Shared Instrumentation Laboratory, Room 016 | Monday, April 13, 2026

This is the master document — read first

Monday's session uses THREE instruments back-to-back on the same six samples: UV-2550 → FluoroMax-3 → Lambda 750. You prepare the samples ONCE (Sunday evening) and run them THREE times (Monday morning).

This document contains the full sample preparation (Part A) and the UV-2550 procedure. The other two documents (FluoroMax-3, Lambda 750) cross-reference back to Part A of this one. Bring all three printed protocols Monday morning.

Recommended reading order the night before: this document cover-to-cover, then skim the FluoroMax-3 and Lambda 750 documents so you know what's coming.

Session sequence — three instruments, one set of samples

1. UV-2550 (this document) — 09:30 – 10:15. Fast absorption scans 200 – 800 nm. Tells you WHERE each molecule absorbs (λ_max) and HOW concentrated it is (A value). Output feeds directly into the next two instruments: λ_max = excitation wavelength for FluoroMax; A value → dilution factor for FluoroMax.

2. FluoroMax-3 (document 2) — 10:30 – 11:45. Fluorescence excitation and emission scans. Uses the UV-Vis λ_max values as the excitation wavelengths. Samples are diluted by D = A_measured / 0.05 before scanning.

3. Lambda 750 (document 3) — 12:00 – 13:00. High-resolution UV/Vis/NIR double-beam rescan. Two jobs: (a) cross-validate Monday's UV-2550 spectra against a research-grade instrument, and (b) extend the scan range into the NIR (800 – 2500 nm) to see water/ethanol O-H overtones that the UV-2550 cannot reach.

Total wall-clock time including setup, warmups, walks between rooms, and packout: ~4 hours.

UV-2550 session goal

Collect UV-Vis absorption spectra for six fluorophores (quinine, yellow highlighter, pink highlighter, curcumin, chlorophyll/tea, salicylate).

The UV-Vis scan tells us (a) exactly where each molecule absorbs light — this becomes the excitation wavelength for the FluoroMax session, and (b) whether the concentration is in the correct range (A < 0.1 at λ_ex means no inner-filter effect when we move to fluorescence).

Expected duration: 45 – 60 minutes (UV-Vis is much faster than fluorescence).

Key concept — Beer-Lambert law

$A = \varepsilon \cdot b \cdot c$

Where A = absorbance (dimensionless), ε = molar absorptivity (M^{-1} cm^{-1}, characteristic of the molecule at a given wavelength), b = path length (1 cm for your cuvettes), c = concentration (M).

What this means in practice: the absorbance value on the screen is directly proportional to the amount of molecule in the cuvette. Double the concentration → double the A. For reliable data, aim for $0.1 \leq A \leq 1.0$ at the peak. Above 1.5 the detector begins to lie; below 0.05 the signal disappears into noise.

Safety checklist (applies to all three instruments)

- Safety glasses, lab coat, nitrile gloves — keep these on all session long.
- Ethanol is flammable; caps closed between uses. No open flame, no hot plates nearby.
- UV-2550 / Lambda 750 deuterium lamps are UV sources — do not stare into the sample compartment with the lid open.
- FluoroMax-3 xenon arc lamp is a very bright UV source — never look into the sample chamber with the shutter open.
- Locate sink, eyewash, and safety shower in each room before starting.
- All liquid waste into your labeled waste bottle; carry it out (no storage in 016).

Part A — Sample preparation at home (Sunday evening)

Do this the night before. All six samples take about 30 minutes start to finish. Label every bottle with the sample name and date. Store overnight at room temperature, out of direct sunlight. These same samples are used for UV-2550, FluoroMax-3, AND Lambda 750 — you are prepping once and running three times.

Things you already own that you will use

- 12-pack of 4 oz amber glass bottles with caps (Piniwon)
- 100 disposable 3 mL plastic dropper pipettes (Moveland)
- 95% ethyl alcohol (Curated Chemical)
- Distilled water (Dura Cleanse, 20 oz bottles)
- Tonic water, yellow + pink highlighters, turmeric, green tea bags, aspirin 325 mg, baking soda
- Coffee filters (kitchen) or unused Kimwipes for filtration
- A clean measuring cup or shot glass for rough 20 mL volumes

Sample 1 — Quinine stock (from tonic water)

1. Open a fresh bottle of tonic water. Pour ~60 mL into a clean glass and let it sit uncovered for 10 minutes to release most of the dissolved CO2. Bubbles scatter light and will ruin your fluorescence emission scan.
2. Decant the de-gassed tonic water into an amber bottle. Label: "Quinine stock (tonic water, degassed)".
3. Do NOT dilute. Tonic water already contains ~80 mg/L quinine, which is near the sweet spot for both UV-Vis and fluorescence.

Sample 2 — Yellow highlighter ink (fluorescein-family)

4. Pop the back cap off a yellow Amazon Basics highlighter. Pull out the ink reservoir (a white fibrous stick saturated with dye).
5. Place it in a small cup with 30 mL of distilled water. Squeeze and swish gently with a pipette for 30 seconds until the water turns deep yellow-orange.
6. Filter through a coffee filter into an amber bottle to remove fibers. Label: "Yellow HL stock".
7. For UV-Vis, start with 1 drop of stock in 3 mL distilled water (done at the lab). For FluoroMax, dilute further per the formula D = A_measured / 0.05 computed during the UV-Vis session.

Sample 3 — Pink/orange highlighter ink (rhodamine-family)

8. Same procedure as Sample 2, using a pink or orange highlighter and a fresh 30 mL of water. Label: "Pink HL stock".
9. Same dilution strategy: 1 drop into 3 mL for UV-Vis, then additional dilution for FluoroMax based on measured A.

Sample 4 — Curcumin (from turmeric in ethanol)

10. Measure ~1/4 teaspoon (~1 g) of ground turmeric into a clean glass.
11. Add 20 mL of 95% ethanol. Stir vigorously for 30 seconds with a plastic pipette. Let stand for 5 minutes; the solids will settle.
12. Carefully decant the yellow-orange supernatant through a coffee filter into an amber bottle. Do not transfer the settled solids.
13. Label: "Curcumin stock (turmeric/EtOH)". For UV-Vis start with neat stock; for FluoroMax dilute per measured A.

Sample 5 — Chlorophyll / catechins (from green tea in ethanol)

14. Cut open one Soeos green tea bag and empty the leaves into a small glass.
15. Add 20 mL of 95% ethanol, not hot water. Stir and let stand for 10 minutes. The ethanol extracts chlorophyll and polyphenols.
16. Filter the green-brown liquid through a coffee filter into an amber bottle. Label: "Green tea EtOH extract".
17. For UV-Vis start with neat stock; for FluoroMax dilute per measured A.

Sample 6 — Salicylate (from aspirin + baking soda)

18. Crush one 325 mg aspirin tablet with the back of a spoon on a clean dish.
19. Transfer to a glass and add 20 mL of distilled water plus a small pinch (~100 mg, the size of a match head) of baking soda. The bicarbonate helps hydrolyze acetylsalicylic acid to salicylate, which is the actual fluorophore.
20. Stir for 1 minute. The binder from the tablet will not fully dissolve. Filter through a coffee filter into an amber bottle.
21. Label: "Salicylate stock (aspirin/NaHCO3)". Scan neat for both UV-Vis and fluorescence.

Blank — pure solvents

22. Pour ~30 mL of distilled water into one amber bottle. Label: "Blank — H2O".
23. Pour ~30 mL of 95% ethanol into another. Label: "Blank — EtOH". You will need BOTH blanks on all three instruments (aqueous samples use the water blank; ethanol samples use the ethanol blank).

Part B — Packing list for Monday morning (master list)

Everything you need for all three instruments goes into one backpack the night before. 016 does not permit storage — everything in, everything out.

Category	Item	Qty
Cuvettes	Fluorescence quartz cuvettes, 10 mm, 4 clear sides (Teflon lidded)	2

Category	Item	Qty
	UV-Vis quartz cuvettes (Lifestyle Visions 4-pack, as backups)	2
Cleaning	Kimwipes or lint-free lens tissue	1 box
	Ceramic tweezers (for holding cuvettes)	1
Solvents	Wash bottle filled with distilled water	1
	Wash bottle filled with 95% ethanol	1
	Empty waste bottle (Piniwon amber, clearly labeled WASTE)	1
Samples	All six labeled amber sample bottles from Part A	6
	Two labeled blank bottles (H2O, EtOH)	2
Transfer	Disposable 3 mL plastic pipettes	~20
	Small 5 mL plastic bottles for bench dilutions	6
PPE	Safety glasses, nitrile gloves, lab coat	as needed
Records	All three printed protocols + pen	1 set
	USB stick (for copying data off each instrument PC)	1
	Phone for photos of samples and screens	1

Part C — First 15 minutes at the UV-2550

The UV-2550 warmup is faster than the FluoroMax because it has simple lamps (no xenon arc), but it still wants ~10 – 15 minutes for the D2 lamp to stabilize.

Step C1 — Power on

24. Turn on the instrument's main power switch.
25. Wait. You will hear the lamps click on. The front panel display should show a self-test sequence ending in READY.
26. Start UVProbe (the Shimadzu software) on the control PC. It will perform its own instrument initialization; let it finish.
27. Start a 15-minute warmup timer. Use the time for cuvette cleaning and the first bench dilutions.

Step C2 — Cuvette preparation

The same cuvette cleaning ritual applies to all three instruments. Fingerprints on the optical faces at 250 nm will absorb more strongly than a dilute sample.

28. Hold the cuvette by its frosted top edge with ceramic tweezers, never with bare fingers on the optical faces.
29. Rinse interior 3× with distilled water, 1× with 95% ethanol, 1× with distilled water.
30. Drain on a Kimwipe; polish the optical faces with a fresh Kimwipe, one wipe per face.
31. Inspect under room light. No streaks, no lint, no droplets.
32. Dedicate one cuvette to "aqueous / standards" and the other to "dyes" (highlighters + turmeric).

Step C3 — Software configuration

Parameter	Value	Why
Measurement mode	Spectrum	Wavelength scan, not fixed-λ kinetics
Measurement range	200 – 800 nm	Covers deep UV through visible
Scan speed	Medium (~300 nm/min)	Good SNR without being slow
Sampling interval	1.0 nm	Standard spectral resolution
Slit width	2.0 nm	UV-2550 default
Photometric mode	Absorbance (A)	Log scale, not %T
Data export	CSV per spectrum	Loadable in Python later

Step C4 — Baseline correction

UV-Vis is a ratio measurement. You need to tell the instrument what "empty" means before it can tell you what "sample" means.

33. Fill BOTH cuvettes with the same solvent blank (start with distilled water).
34. Place one cuvette in the sample position (front) and one in the reference position (back). Close the lid.
35. In UVProbe, click "Baseline" or "Auto Zero" and run a full 200 – 800 nm baseline scan. The result should be a flat line at A ≈ 0 with tiny noise. If it wanders more than ±0.02, re-clean both cuvettes and redo.
36. Save the baseline. The software will subtract it from every subsequent sample scan automatically until you change solvents.
37. You will need to re-baseline once when you switch from water samples to ethanol samples. Plan: do water samples first (quinine, yellow HL, pink HL, salicylate), then ethanol samples (curcumin, green tea).

Part D — Per-sample scan plan

For each sample you run just ONE scan: a full 200 – 800 nm wavelength sweep. That single scan gives you every absorption peak of the molecule.

Order of samples (water samples first, then re-baseline, then ethanol samples):

38. Blank check (water, both cuvettes) — confirm baseline is still flat
39. Sample 1 — Quinine (neat tonic water)
40. Sample 2 — Yellow HL (1 drop / 3 mL water)
41. Sample 3 — Pink HL (1 drop / 3 mL water)
42. Sample 6 — Salicylate (neat stock)
43. Re-baseline with ethanol in both cuvettes
44. Sample 4 — Curcumin (neat or diluted)
45. Sample 5 — Green tea extract (neat or diluted)

Expected peaks by sample

Sample	Solvent	Expected A peaks (nm)	Target A
1. Quinine	Water (tonic)	~225, ~275, ~335	0.3 – 0.8
2. Yellow HL (fluorescein)	Water	~230, ~490	0.3 – 0.8
3. Pink HL (rhodamine)	Water	~230, ~554	0.3 – 0.8
6. Salicylate	Water	~230, ~296	0.3 – 0.8
4. Curcumin	Ethanol	~265, ~425	0.3 – 0.8
5. Green tea extract	Ethanol	~270, ~410, ~665	0.3 – 0.8

Read off the absorbance at the strongest peak. Write it down. This is the number you will use in Part F to calculate the dilution factor for the FluoroMax session.

Part E — Standard procedure for every sample

A 5-step cycle per sample. At ~45 seconds per scan, the whole session is a fast ~45 minutes.

46. Rinse the sample cuvette once with ~1 mL of the new sample (pre-rinse). Discard into waste bottle.
47. Fill the sample cuvette with ~3 mL of sample. Leave the reference cuvette filled with the most recent blank.
48. Wipe both polished faces of the sample cuvette with a fresh Kimwipe. Inspect. Place in the sample position.
49. Run "Spectrum" scan. The scan takes ~45 seconds. Watch the live plot.
50. Save/export as CSV with file name: 20260413_UVVis_S{n}_{sampleName}.csv

Reading the screen as it scans

- If A_max < 0.1: sample is too dilute. Dilute less next time, or scan neat stock.

- If A_max > 1.5: sample is too concentrated. Dilute 10× and rescan.
- If the whole spectrum is above A = 1 with no peaks: cuvette is dirty, or the sample is cloudy. Refilter.
- If you see sharp spikes at 656 nm or 486 nm in aqueous samples: hydrogen Balmer lines from the D2 lamp leaking through. Ignore; they are not real.

Part F — Calculating the FluoroMax dilution (handoff)

This is the handoff to the fluorescence session. After you have all six UV-Vis spectra, use this quick formula for each sample:

Dilution factor for fluorescence

D = A_measured / 0.05

Where A_measured is the absorbance at the peak you intend to EXCITE at in the FluoroMax. The target of 0.05 keeps the sample safely below the inner-filter threshold (A < 0.1) at the excitation wavelength.

Example: yellow HL shows A = 0.7 at 490 nm. D = 0.7 / 0.05 = 14. Dilute 1 drop of current sample into 14 drops of water for the FluoroMax cuvette.

Example: curcumin shows A = 0.4 at 425 nm. D = 8. Dilute 1 drop into 8 drops of ethanol.

For very dilute samples (A < 0.1 already): no dilution needed — use as-is for fluorescence.

Write the computed dilution factor next to each sample in your notebook. Carry the small 5 mL plastic bottles with the diluted aliquots to the FluoroMax room.

Handoff to Lambda 750

The Lambda 750 session uses the SAME moderately concentrated samples as the UV-2550 (no further dilution needed). Keep the undiluted amber bottles aside for the third instrument. You will also scan the H2O and EtOH blanks extended all the way to 2500 nm on the Lambda 750 to see water and ethanol NIR overtones.

Part G — UV-2550 shutdown (and handoff)

51. In UVProbe, verify all six .csv files are saved. Open two at random to confirm they parse.
52. Copy the session folder to the USB stick.
53. Close UVProbe.
54. Turn off the UV-2550 at the main power switch.
55. Wipe the bench. Do NOT pack out yet if you're continuing to FluoroMax-3 or Lambda 750 in the same room.

If you are continuing straight into the FluoroMax-3 session, walk over to the FluoroMax and turn it on NOW so the xenon lamp can warm up (20 minutes) while you do the final dilution calculations at the bench.

Part H — Data management

56. Copy the session folder into GITHUB/science/research/20260413 UV-Vis/raw/
57. Create an index.md with: date, instrument (Shimadzu UV-2550), operators, samples, file list, and A_max values for each sample.
58. Load one CSV in Python (pandas + matplotlib) and save a reference plot as PNG.
59. Commit: "Add UV-2550 first session — 6 fluorophores, absorption spectra".

60. Bonus: add a small table linking each UV-Vis peak to the corresponding FluoroMax excitation wavelength and Lambda 750 NIR overtone — that cross-instrument table is the scientifically interesting result.

Part I — Troubleshooting cheat sheet

Symptom	What to try
Baseline is not flat (wavy, tilted, wandering)	Re-clean both cuvettes. Confirm both contain the SAME blank solvent. Re-run baseline.
All samples read A ≈ 0	Sample cuvette in wrong slot; or both cuvettes accidentally still contain blank; or wavelength range excludes the real peak. Check.
All samples read A > 3 (flat)	Sample way too concentrated. Dilute 100×.
Spectrum has sharp noise spikes	Deuterium lamp emission lines leaking through — harmless, ignore. Or: cuvette has a bubble; tap it out.
Peak at 200 – 210 nm only, nothing else	Solvent or dissolved oxygen absorption, not your molecule. Scan range too tight; widen it.
Peak position does not match literature	Different solvent shifts peaks. Curcumin in ethanol ≠ curcumin in water. Your numbers are fine.
Absorbance drifts during a scan	Sample evaporating (ethanol) — cap with Teflon lid between scans. Or sample degrading in UV — scan faster.
Quinine peak at 335 nm is weak	Tonic water is dilute (~80 mg/L quinine). Peak will be small but clearly present. If missing, the tonic water is old.

Part J — Post-session reflection

Questions to discuss with Vivian before the FluoroMax session:

- Which sample has the largest ε (molar absorptivity)? How do you know from the A values?
- For the samples that show TWO peaks (one in UV near 230 nm, one in the visible), what do the two peaks represent? (Hint: different electronic transitions — π→π* and n→π*.)
- Why does green tea extract have peaks in both the blue (~410 nm) and the red (~665 nm)? What does that tell you about chlorophyll's light-harvesting strategy?
- If you triple the concentration of curcumin, what should A at 425 nm become? (Beer-Lambert predicts exactly tripling.)
- Compare your measured λ_max for each sample against the literature values. Within 5 nm is excellent; within 10 nm is normal.

Looking ahead: for each sample, compare where it ABSORBS (UV-Vis peak, this session) to where it EMITS (FluoroMax peak, next session) and where its NIR vibrational overtones show up (Lambda 750, session 3). The gap between absorption and emission is the Stokes shift. The NIR overtones tell you about the chemical bonds in the SOLVENT, not the dye.

— End of UV-2550 protocol — continue to FluoroMax-3 (document 2) —